

Delta Robot

Hardware & Performance Specification

Kinematics

Parameter	Symbol	Value
Type	—	Parallel delta, 3-DOF
Base triangle radius	f	157 mm
End-effector radius	e	35 mm
Bicep / upper arm	rf	200 mm
Forearm / lower arm	re	400 mm

Workspace

Axis	Limit
X	± 151.6 mm
Y	± 151.6 mm
Z	−196.9 mm to −500.0 mm
Home position	(0, 0, −350) mm

Joint Limits

Joint	Min	Max
θ1	0°	82.1°
θ2	0°	85.5°
θ3	0°	85.5°

Motors (× 3)

Parameter	Value
Model	RobStride RS-00
CAN IDs	1, 2, 3 (29-bit extended)
Max speed	50 rad/s
Control mode	Profile Position (PP)
Position tolerance	2.0 mm (FK verified)

Gripper

Parameter	Value
Type	Pneumatic solenoid valve
CAN ID	4 (11-bit standard)
Close settle time	0.5 s
Open settle time	0.3 s

CAN Bus

Parameter	Value
Interface	can0 (SocketCAN)
Bitrate	1 Mbit/s
Motor IDs	1 – 3
Solenoid ID	4

Camera

Parameter	Value
Model	Intel RealSense D455
Baseline	95 mm
Shutter	Global
Offset from base	X = 97.6 mm, Y = 11.9 mm, Z = 432.0 mm

Speed Presets

Mode	Cartesian speed	Motor velocity	Motor acceleration	Settle delay
low	80 mm/s	1.0 rad/s	2.0 rad/s ²	0.30 s
medium	2000 mm/s	25.0 rad/s	50.0 rad/s ²	0.15 s
max	4000 mm/s	50.0 rad/s	100.0 rad/s ²	0.10 s